

JOSEPH GITAU CHEGE

Software Engineer — Backend Systems, Identity & Notification Platforms, Cloud Reliability

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PROFESSIONAL SUMMARY

Software engineer with 3+ years building and operating production backend systems at scale in C#, Python, C++, and JavaScript/TypeScript. Proven track record designing and shipping notification and messaging capabilities for an institutional user base, including real-time push, in-app, and multi-channel delivery, alongside password and biometric authentication flows. Delivered 99.9% uptime on mission-critical services through disciplined incident response, live-site monitoring, and post-incident improvement, in line with enterprise-grade SLAs. Strong background in Azure, Docker, Kubernetes, and queue-based, high-availability architectures. Outside of professional work, applies the same systems-thinking across a wide personal project portfolio spanning media platforms, embedded/hardware-adjacent software, mobile apps, and game development — building product-minded, end-to-end software that runs everywhere from the cloud down to microcontrollers and USB peripherals. Brings an AI-first mindset with a focus on reliability, observability, security, and continuous modernisation, and thrives collaborating across cross-functional and cross-team engineering groups.

CORE COMPETENCIES

Identity, Notification & Messaging: Notification delivery services (push, in-app, multi-channel) · Authentication flows (password, biometric) · Real-time messaging & presence · Security signals & account lifecycle concepts

Backend & Distributed Systems: C#, Python, JavaScript/TypeScript, C++ · API & backend service development · Queue-based architectures · Scalable, high-availability service design · Database design (PostgreSQL, MySQL, MongoDB, Firebase)

Cloud & DevOps: Azure · Docker · Kubernetes · CI/CD pipelines · Bash automation · Proxmox · Resource & cost optimisation

Reliability & Live-Site Operations: Incident triage & escalation · Post-incident reviews · 99.9% uptime delivery · Monitoring & health checks · Enterprise-grade SLAs · SOPs & documentation · Cross-team collaboration · AI-first problem solving

Systems Thinking Across the Stack: Embedded/hardware-adjacent software (Arduino, USB & serial protocols, GPU firmware) · Cross-platform app development (web, desktop, mobile) · Product-minded engineering & self-directed project delivery · Real-time/interactive systems (game telemetry, audio-reactive control, game development)

PROFESSIONAL EXPERIENCE

Engineering Team Lead & Head of IT | Steadfast Academy

April 2025 – Present | Nairobi, Kenya

Notification & Identity Infrastructure

- Designed and built notification delivery capabilities within an in-house ERP system, enabling real-time push notifications, in-app alerts, and multi-channel messaging across mobile and web — directly analogous to high-scale notification service architectures, and covering event triggers, delivery confirmation, and channel fallback logic.
- Implemented authentication flows including password-based login and on-device biometric verification, securing access to sensitive institutional data and demonstrating strong understanding of identity-layer, session management, and account security concerns.
- Built a real-time chat and messaging subsystem within the ERP, handling message delivery, presence, and read-receipt patterns at institutional scale, with attention to delivery reliability under variable network conditions.

Platform & Backend Engineering

- Architected and developed a full-stack ERP system consolidating student records, attendance, finance, and communication workflows, delivering production-ready features end-to-end from design through implementation and validation.
- Wrote Bash automation scripts and CI/CD pipelines that reduced deployment errors by 45% and reclaimed 15+ hours per week.

- Managed Azure cloud infrastructure end-to-end, optimising resource allocation and cutting cloud operating costs by 20%.

Reliability, Incident Response & Live-Site Operations

- Delivered 99.9% uptime on mission-critical systems through proactive monitoring, scheduled maintenance, and disciplined, fully documented incident response — directly aligned with live-site, SLA-driven reliability engineering, including root-cause analysis and follow-up action tracking after every incident.
- Deployed and maintained containerised services using Docker and Kubernetes, improving system uptime by 50% and reducing latency by 30%.
- Designed a high-availability storage architecture on TrueNAS supporting 10TB+ of institutional data, applying redundancy and failover patterns.
- Administered Ubuntu Server and Debian environments — service configuration, log analysis, and OS-level troubleshooting across production systems.

Leadership & Cross-Team Collaboration

- Led a cross-functional engineering team and served as the primary technical point of contact between engineering and institutional stakeholders, collaborating across teams to align on requirements and adoption.
- Spec'ed, built, and maintained the academy's on-premises server and network infrastructure from the ground up, including structured cabling, switch configuration, and asset tracking.

Software Consultant | Gituamba Girls

Jan 2024 – Mar 2025 | Kenya

- Designed and built a full-stack school management system as a solo developer (C#, Python), covering student records, finance, attendance, and academic reporting — owning delivery end-to-end from design through deployment.
- Developed RESTful APIs and database schemas (PostgreSQL/MySQL) serving daily school operations.
- Wrote automation scripts that reduced manual administrative work by 4 hours/week.
- Managed deployment and maintenance on cloud/on-prem infrastructure end-to-end.

Software Engineering Intern | VisionFund Kenya (MicroFinance)

Oct 2023 – Dec 2023 | Kenya

- Built internal automation tools in C# and Node.js and refactored MySQL database schemas, cutting query response times by 15% on critical financial reporting systems.
- Contributed to backend maintenance and debugging on live production financial systems, developing habits of careful, low-risk, well-documented changes within a regulated, high-accountability environment.

SELECTED PROFESSIONAL PROJECTS

In-House ERP — Steadfast Academy (2024–2025)

- Designed and shipped a full-stack ERP application end-to-end — from requirements through deployment — consolidating student, attendance, finance, and communication records, with notification and messaging capabilities serving the entire institution.

Notification & Authentication Infrastructure — Steadfast Academy (2025)

- Designed and implemented real-time notification delivery (push, in-app) and authentication (password + biometric) subsystems within the ERP mobile and web platform, serving hundreds of daily active users.

BetterFarm — AgriTech Platform, Personal Project (2024)

- Lead architect for a cloud-native backend system on Azure; integrated real-time data pipelines and deployed scalable backend services using high-availability design principles.

PERSONAL PROJECTS & OPEN SOURCE

A self-directed portfolio reflecting product-minded, full-stack engineering beyond the day job — spanning web, desktop, mobile, embedded, and game systems, and consistently applying the same reliability and systems-design habits used in professional backend work.

Media & Entertainment Platforms

- Jellyfin Music Streaming Player (Next.js) — Built a full-featured web client for self-hosted music libraries, with real-time playback queueing and a modern streaming-app UX on top of the open-source Jellyfin API.
- Jellyfin "Yearly Wrapped" Stats App — Designed a Spotify-Wrapped-style analytics app that aggregates and visualizes a year of listening history from self-hosted media data.
- Electron Desktop Jellyfin Client — Packaged a web media player into a cross-platform native desktop app, adding system tray, media-key, and OS-level integration.

Hardware, RGB & Embedded Systems

- Game-Telemetry LED Lighting Bridges (Forza, Delta Force) — Built custom software bridges that translate live in-game telemetry into OpenRGB-controlled lighting, mapping software state to physical hardware feedback in real time.
- OpenRGB ↔ SignalRGB Interoperability Bridge (Python) — Wrote a protocol-translation layer connecting two otherwise incompatible RGB device ecosystems, enabling cross-vendor lighting control.
- Sound-Reactive LED Companion App — Built a mobile app that analyzes live audio input and drives synchronized LED effects on connected hardware in real time.

Mobile Applications

- React Native IEM Equalizer — Engineered a mobile audio equalizer with automatic USB-C DAC detection, applying real-time signal processing to external audio hardware.
- Farming Companion App — Built a mobile app with map-based field mapping for tracking agricultural plots, activities, and yields.
- Personal Finance Tracker — Designed a mobile budgeting and expense-tracking app with category-based spending insights.

Game Development

- Unity FPS Game — Designed and built a first-person shooter prototype in Unity/C#, covering gameplay mechanics, physics, level design, and player systems.

Home Automation & IoT

- Home Automation Monorepo (14+ sub-projects) — Architected and maintain a multi-project repository spanning Arduino-based fan controllers, IoT lighting bridges, and automation bots, applying consistent tooling and structure across a diverse hardware/software portfolio.

Security, Surveillance & Data Visualization

- Hikvision Device Bridge & Access Log Monitor — Built an integration bridge for Hikvision surveillance hardware alongside a real-time access-control log monitoring tool.
- 3D Family Tree Visualization — Built an interactive 3D genealogy visualization tool for exploring multi-generational family relationships.

TECHNICAL SKILLS

Languages: C#, Python, JavaScript/TypeScript(React, Nextjs Node), C++, Bash, Kotlin, Java Spring Boot.

Backend & Data: PostgreSQL, MySQL, MongoDB, Firebase, API development, schema design, query optimisation, notification systems, messaging infrastructure, authentication flows (password, biometric)

Cloud & DevOps: Azure (primary), GCP (upskilling), Docker, Kubernetes, Proxmox, CI/CD pipelines

Reliability & Ops: Linux (Ubuntu/Debian), incident triage & escalation, monitoring, high-availability design, enterprise SLAs, documentation & SOPs

Networking & Hardware: LAN/WAN, VLAN, patch panels, firewall configuration, server hardware, structured cabling

Embedded, Hardware & Cross-Platform: Arduino, USB/serial device integration, OpenRGB SDK, GPU firmware/BIOS internals, Electron, React Native, Unity/game development

EDUCATION

B.Sc. Computer Science — Umma University, Nairobi

2021 – 2024

Key modules: Data Structures & Algorithms, Database Management Systems, Network Security, Operating Systems, Computer Architecture.